

Using Hierarchical Models to Estimate Heterogeneous Effects

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Abstract

Conditional theories are ubiquitous in political science, but standard interaction models cannot handle this proliferation of theoretically informed modifiers. This paper describes why, when, and how to use hierarchical models to encompass multiple sources of heterogeneity and regularize estimates. Doing so is simple, and starts with identifying the key sources of heterogeneity and grouping observations accordingly. Next, fit a hierarchical model with two linked regressions, one connecting treatment with the outcome, and another that models the treatment effects with potential sources of heterogeneity and partially pools group estimates. This captures systematic and random variation in heterogeneous effects, encompasses many interactions, and fits commonly used modeling frameworks, but regularization in this model breaks down if there are many small groups. I document these claims with a simulation analysis and reanalysis of a published study.

*Thanks to Taylor Kinsley Chewning, Andrew Gelman, Robert Kubinec, Carlisle Rainey and Rod Sturdivant for helpful comments.

Social scientists often propose conditional theories (Clark and Golder, 2023) and are interested in how different units respond to the same stimulus for theoretical, normative or policy reasons. While most theories consider a single modifier, many theories propose distinct modifiers for the same independent variable. Therefore, multiple modifiers are common. For example, international relations experiments on audience costs (Tomz, 2007) have considered how foreign policy dispositions (Kertzer and Brutger, 2016), partisanship (Levendusky and Horowitz, 2012), gender (Barnhart et al., 2020; Schwartz and Blair, 2020) and policy preferences (Chaudoin, 2014) modify reactions to a leader backing down from a threat. Other audience cost experiments manipulate casualties (Davies and Johns, 2013), how the leader backs down (Lin-Greenberg, 2019), other elite reactions (Levendusky and Horowitz, 2012) and promises vs threats (Levy et al., 2015).

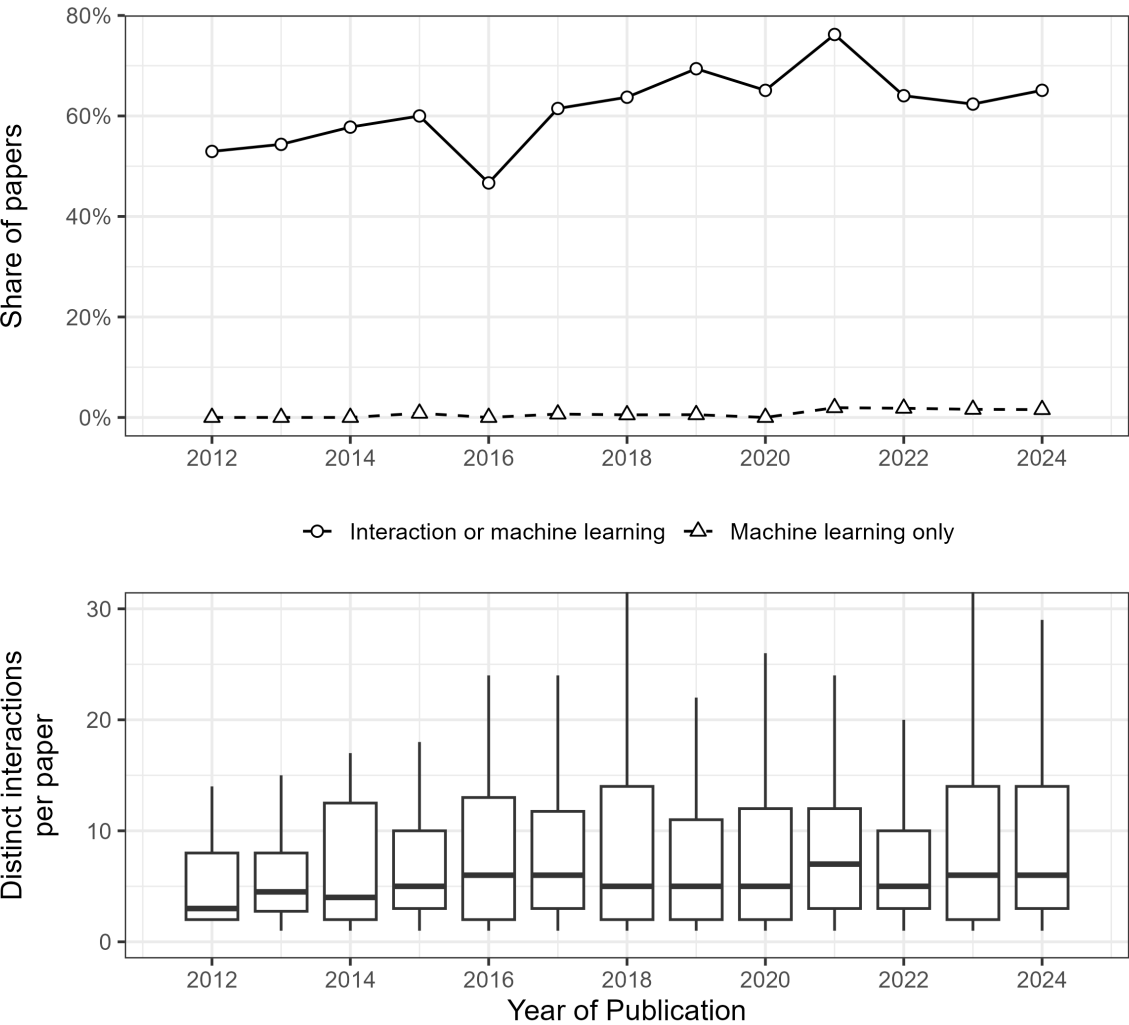
Many potential modifiers are evident in how scholars conduct quantitative research. In the *American Political Science Review*, *American Journal of Political Science* and *Journal of Politics*, many researchers estimate some modifier of treatment effects, either to test their argument or explore others. Figure 1 documents the prevalence of heterogeneous effects estimates from interactions or machine learning in Harvard Dataverse replication data from the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics* from 2012 to 2024. In this figure I count the number of papers that fit at least one interaction or machine-learning model of effect heterogeneity in their replication data to show the prevalence of these tools. I did this by downloading all quantitative replication materials hosted in the Harvard dataverse, and then measuring the prevalence of heterogeneous effects estimates. I then used an open-weights LLM (gpt-oss-120b) to parse the code and identify models and interactions.¹

Interactions are very common. In most years, 60% of papers in the APSR, AJPS, and JOP estimate at least one interaction. Use of machine-learning tools for heterogeneous effects is much less common, and is usually in the single-digits.²

¹Blind hand validation of a stratified sample puts precision above 90%, with details in the appendix.

²As another indicator of interaction popularity, Brambor, Clark and Golder (2006) has 8,279 citations as of

Heterogeneous Effect Estimation in Leading Journals
2012-2024



Lower panel covers papers estimating at least one interaction and drops outliers from view

Figure 1. *Prevalence of heterogeneous effect estimates, 2012 to 2024.*

Moreover, researchers fit many interactions, as the median paper estimates 6 distinct interaction terms. A few outliers estimate many more. Most applied research relies on parametric interactions, with different theories and robustness checks leading to multiple interactions in most papers.

Unfortunately, the workhorse regression interactions that many scholars rely on are ill-suited to test the growing number of conditional arguments in social science. Scholars cannot ignore heterogeneity, but the most common tools either increase the risk of spurious results or are hard to interpret and use. Interaction terms and subgroup analysis are ubiquitous because they are relatively easy to interpret, but they increase the risk of spurious results. Many political science analyses have low power even to detect effects without interactions (Arel-Bundock et al., 2026). Adequate power to estimate even a single interaction can require significantly more data (Gelman, 2018), which may be prohibitively expensive or impossible. As a result, statistically significant heterogeneous effect estimates may be far too large—the result of noise in the data, not systematic differences. This problem is partially responsible for widespread issues replicating findings based on interactions (Simmons, Nelson and Simonsohn, 2011). Adding more modifiers makes interpretation harder and increases these power problems. Researchers might instead fit separate pairwise interactions for each modifier, but doing so multiplies models and corresponding multiple-testing problems.

Given multiple sources of heterogeneity, machine-learning tools such as random forests (Green and Kern, 2012; Wager and Athey, 2018), support vector machines (Imai and Ratkovic, 2013), and ensemble methods (Grimmer, Messing and Westwood, 2017; Künzel et al., 2019; Dorie et al., 2022) are more likely to avoid over-fitting. These machine learning algorithms usually have some regularization component and can discover complex patterns and high-dimensional variation across multiple modifiers.³ These tools can be difficult to interpret and

September 2026.

³LASSO models sometimes address similar issues (Ratkovic and Tingley, 2017). Blackwell and Olson (2022) describe a lasso approach to interactions of a single modifier and all the variables in a model that sits between

implement, however, especially in smaller social science datasets. The relative lack of theoretical guidance and interpretability is especially problematic for testing the relative weight of multiple conditional arguments. If researchers want to compare the impact of different modifiers, machine learning models use quantities that many applied researchers are less familiar with, so these techniques are far less popular than interactions.

Despite a large statistics literature documenting how regression-based interpretation and regularization make Bayesian hierarchical models a useful tool for heterogeneous effects estimation (e.g., Feller and Gelman (2015); McElreath (2016); Dorie et al. (2022)), political scientists rarely use this tool. For instance, of the three applied political science citations of Feller and Gelman (2015), only Marquardt (2022) models treatment effects. The MAIHDA (Multilevel Analysis of Individual Heterogeneity and Discriminatory Accuracy) framework in intersectionality research is a noteworthy exception (e.g. Much, 2025). These researchers estimate distinct effects for groups defined by class, race and gender, and partial pooling helps give reasonable estimates for small subgroups in survey data. I use a similar logic to estimate how treatment effects vary across groups.

This paper provides practical guidance by explaining how, when and why to use hierarchical models of heterogeneous effects. The hierarchical varying slopes strategy I advocate here addresses the power shortcomings of interactions while allowing more theoretical guidance and comparison than machine learning. It does so by regularizing estimates of multiple modifiers, thus trading some bias of specific effects for reduced variance in effect estimates. Taking this side of the bias-variance tradeoff improves accuracy overall, and reduces errors about the sign and magnitude of non-zero effects.

Using hierarchical models is more flexible than standard interactions but easier to implement and interpret than machine learning. It preserves a straightforward structure while ac-

machine learning and linear regressions. This considers a single interaction of interest and bias from other unmodeled interactions with controls.

commodating more factors and ameliorating the downsides of subgroup analysis. The main weaknesses are that unlike machine learning, the hierarchical approach lacks the flexibility to discover high-dimensional heterogeneity and regularization will break down with many small groups. It can also be slow to fit on large datasets, depending on the underlying sampler.⁴

There are two key steps when theory and data make using hierarchical models worthwhile. First, researchers should identify potential modifiers of a treatment effect and use them to group observations and model treatment effects. Second, they should take that model of treatment effects and connect it to a model linking the treatment and outcome, implementing a type of varying slopes model. Modeling heterogeneous effects produces interpretable results, which helps researchers compare different modifiers. It also allows researchers to compare different sources of heterogeneous effects and describe effect variation. These are crucial advantages given many conditional theories.⁵

While frequentist estimation of hierarchical models is possible, Bayesian estimation is easy, more informative and more robust (Stegmueller, 2013). Bayesian estimation provides crucial information by connecting parameters through common prior distributions, thereby regularizing estimates and propagating uncertainty. Working with posterior distributions also gives researchers more flexibility to describe how and when effects vary and use prior predictive checks to examine the assumptions in their models. While computation and coding were once a barrier to employing Bayesian methods, fitting a wide range of hierarchical models is straightforward with the `brms` package in R (Bürkner, 2017).⁶

In the remainder of this paper, I start by describing how and when to estimate hierarchical models of heterogeneous effects. I then use simulations to compare OLS and hierarchical

⁴Goplerud (2021) considers a model that uses Bayesian structured sparsity to estimate which group coefficients are similar and which are different. In this approach, researchers use theory to inform potential groups, but the data determines common estimates for groups.

⁵The downside is that when partial pooling moves subgroup estimates towards the overall mean, testing claims about specific groups is not wise.

⁶I provide example code below and in the appendix.

estimates of treatment effects under different conditions. Finally, I demonstrate the process by analyzing a study by Barber and Pope (2019), who attempt to disentangle partisanship and ideology. The reanalysis documents the importance of jointly estimating modifiers in a single regularized model.

1 Hierarchical Modeling of Heterogeneous Effects

There are two steps in hierarchical models of heterogeneous effects with varying intercepts, varying slopes and group-level predictors of the slopes (Gelman and Hill, 2007). First, researchers must identify potential sources of heterogeneity, and think about the right model of heterogeneity. This will also depend on what variation is most important and interesting. Theory, policy concerns, or normative factors are all possible motivations.

This first step determines what heterogeneous effects a researcher estimates. It is analogous to researchers thinking through a regression specification and requires similar care. Researchers need to define what variation is most important, link heterogeneous effects to theory, and structure modeling. This is especially important for pre-registration, and could reduce the amount of exploratory analyses in registrations. Not thinking carefully about sources of heterogeneity will obfuscate results and can hinder model fitting.

There are three general approaches to defining key modifiers. First, researchers can use combinations of other treatments, especially when an intervention has several dimensions but theory emphasizes one of them. The experimental design determines modifiers, and the model estimates heterogeneous treatment effects. If researchers want to know how different issues shape the impact of elite foreign policy cues (Guisinger and Saunders, 2017), they could include indicators of issues, for instance. A similar application of hierarchical estimators for topic-sampling experiments estimates how a treatment effect varies across topics (Clifford and Rainey, 2024).

At the moment, researchers often estimate the impact of different treatment combinations with fully crossed interactions. This estimates the impact of a treatment across experimental strata, but risks spurious results by functionally estimating subgroup results. These estimates are often noisy, however, and can rely on 50 or fewer data points in smaller samples.

A second approach uses unit, demographic and contextual factors to estimate effect heterogeneity. Here, researchers examine what factors within or around units shape their response to an independent variable. Researchers could use a mix of individual and contextual factors to predict divergent consequences of a survey experiment treatment. Such a model might include factors such as an indicator of state of residence, age, gender, and race.⁷ For example, Alley (2021) uses alliance characteristics to examine when alliance membership increases or decreases military spending.

Third, researchers might use hierarchical models to address specific policy concerns. Policy analysts often want to know how an intervention impacts a specific population. Researchers might want to know if a job-training program improves employment outcomes for black women in the South, for instance. To do this, a researcher might specify a heterogeneous effects model with race, gender and region, plus additional controls or other factors.

After defining moderators and how they relate to treatment effects, the second step is fitting a hierarchical model that links a model of the outcome with a model of heterogeneity. Essentially, researchers model the outcome and the process that produces heterogeneous treatments. The model employs two connected regressions. One regression deals with the outcome, and includes the treatment effects.⁸

The other regression models the treatment effects. In this regression, theory should inform which modifiers predict the treatment effect. To regularize the estimates, this second equation

⁷Extrapolations to a representative sample of a subpopulation might require poststratification.

⁸If other units such as states define the groups, rather than combinations of modifying variables, then adding group-level predictors to this equation is essential. For example, in a model where an effect varies by state, adding state-level variables like ideology, population and GDP would avoid partially pooling small groups too far towards the overall mean.

must include varying intercepts for groups within the data. Groups can be based on combinations of discrete modifiers and binned continuous modifiers. As a result, this model includes both systematic and random variation in the impact of a key independent variable on different groups.

To maximize variation and preserve regularization, researchers should use continuous modifiers directly in the regression and bin them for setting groups. Not binning modifiers will create many small groups, leading the hierarchical component to break down. In the appendix, I document some additional best practices for continuous modifiers, including avoiding smooth priors for ordered effects across bins and the importance of bin boundaries relative to the number of bins.

Setting groups is a critical task that requires balancing detail against regularization. Larger groups can deviate more from the overall mean, as their data contributes more to the posterior than the prior. A few small groups means less need for a hierarchical model. Smaller groups will be shrunk more towards the overall mean. Too many small groups and the regularization component will have less information to work with. Using model fit criteria such as WAIC or leave-one-out cross validation can help guide group formulation. Pre-registration is also crucial, because this is the area with the greatest researcher degrees of freedom.

Should the variables in the heterogeneity equation also appear in the outcome equation? Here, the guidance of Brambor, Clark and Golder (2006) to include constituent terms in an interaction provides a helpful heuristic. If a modifier is correlated with treatment and the outcome, then it is also a useful control, as it has an independent relationship with the dependent variable in addition to modifying the treatment effects. Moreover, if untreated units in every group share the same intercept, the treatment effects absorb baseline outcomes. Including modifiers or group intercepts α_g in the outcome equation prevents this. The functional form of the two equations may differ, however.

I now briefly describe a generic hierarchical model.⁹ For ease of exposition, consider making between-unit comparisons based on an experimental treatment. Start with N units indexed by i , some of which receive a binary treatment T . Assume that the outcome variable y is normally distributed with mean μ_i and standard deviation σ .¹⁰ g indexes researcher-defined groups based on combinations of the modifying variables.

The outcome for each unit depends on a set of group intercepts α_g , an optional matrix of control variables $\mathbf{X}$, and a set of group treatment effects λ_g . The group intercepts are essential: they hold the baseline level of the outcome in each group, so that λ_g captures the effect of the treatment within groups rather than differences between them. When T is binary, estimated λ parameters for untreated units have no impact on the outcome and drop out of the likelihood. For a continuous treatment, effects will depend on the product of T_i and λ .

$$y_i \sim N(\mu_i, \sigma) \quad (\text{Likelihood})$$

$$\mu_i = \alpha_g + \lambda_g T + \mathbf{X}\beta \quad (\text{Outcome Equation})$$

$$\lambda_g = \theta_g + \mathbf{Z}\gamma \quad (\text{Heterogeneity Equation})$$

$$\theta_g \sim N(\mu_\theta, \sigma_\theta) \quad (\text{Group Varying Intercepts})$$

The heterogeneity equation then models those group treatment effects with a systematic and random component. The systematic component is a matrix of predictors $\mathbf{Z}$ and associated parameters γ . $\mathbf{Z}$ can mirror any regression specification researchers might use for an outcome; linear combinations of variables, interactions, smooth functions. Interactions of the modifiers could capture processes where combinations of modifiers produce non-additive jumps in heterogeneity, for instance.

⁹I split the equations for ease of exposition, but it is straightforward to see that this approach places priors on an additive interaction model of the form $y = T * (z_1 + z_2 + z_3 \dots) + \mathbf{X}\beta$

¹⁰Researchers can and should use binary, categorical and other outcome likelihoods.

The random component of the heterogeneity equation is a series of group-specific varying intercepts θ_g . These are critical, because they capture group deviations from the systematic trends expressed in the design matrix $\mathbf{Z}$. Group outliers in the λ estimates are partially pooled back towards the overall mean μ_θ . The variance parameter σ_θ controls the dispersion of the group effects around the overall mean.¹¹

The above model can be fit with Bayesian or frequentist methods, but Bayesian estimation offers important advantages. First, it is more flexible, and including prior information can facilitate model fitting. Priors on the α , β , and γ parameters will matter more in very small datasets. Priors on β should be relatively weak with observational data, because shrinking coefficients on confounders too much might reintroduce omitted variable bias (Hahn et al., 2018). The more important prior is for σ_θ , and depending on beliefs about potential dispersion across groups, half-normal or half student-t priors are common choices. This prior is what helps regularize estimates by pulling extreme groups towards the overall mean μ_θ .

Bayesian modeling offers additional information and a wealth of model-checking tools. STAN and brms, the software I provide guidance for here, facilitates prior predictive checks, where researchers simulate from the model to check what estimates are plausible with specific priors. Working with posterior distributions then provides a wealth of information about effect heterogeneity and propagates uncertainty. Last, Bayesian models can provide useful diagnostics such as divergent transitions that can be warnings of specification problems. It is often better to modify a model with fitting problems instead of brute-forcing it through the sampler.

Depending on the problem, researchers can extend this initial model in many ways. The

¹¹In brms, using non-linear syntax can express a model with a treatment, two controls, and three modifiers as: $y \sim \text{lambda} * \text{treat} + \text{controls}$, $\text{lambda} \sim \text{mod1} + \text{mod2} + \text{mod3} + (1 \mid \text{mod1}:\text{mod2}:\text{mod3})$, $\text{controls} \sim \text{mod1} + \text{mod2} + \text{mod3} + \text{control1} + \text{control2} + (1 \mid \text{mod1}:\text{mod2}:\text{mod3})$, $\text{nl} = \text{TRUE}$. The modifiers and the group intercept must appear in the controls equation as well, because together they form α_g . Omitting them imposes a common baseline on every group, and lambda then absorbs differences in the level of the outcome across groups rather than the effect of the treatment within them.

generative core of Bayesian hierarchical models adds significant flexibility. For instance, they might model the variance of the treatment effects or the outcome. Fitting splines or smooth functions is also possible, or researchers might extend the model in continuous group space with Gaussian process regression (Prati et al., 2023).

In interpreting these estimates, researchers should leverage the full range of information from the different parameters. First, the λ posteriors give the impact of the treatment on each group, and are the core quantity of interest. All λ s reflect a systematic component from the predictors in $\mathbf{Z}\gamma$ and random variation from θ . γ parameters can, depending on the regression, be interpreted as the impact of a change in a modifier on the treatment effects. For example, a γ of .1 on a binary modifier means that λ is .1 higher in expectation when the modifier is one, and .1 lower when it is zero. σ_θ thus measures treatment variation outside the systematic regression. Other techniques such as interactions in OLS with robust standard errors provide less information.

2 Performance on Simulated Data

To assess how this hierarchical model compares to interactions, I first assess their performance on simulated data. For ease of exposition, I consider machine-learning models in the appendix. The appendix compares LASSO, causal forest and BART, estimating the same group effects, and finds that the hierarchical model is usually more accurate.

In the simulations, I fix the number of observations, and then generate different scenarios with distinct groups and distributions of true effects across those groups. All of these scenarios give a group-specific effect of a binary treatment variable. I then estimate those effects with three models. First, ordinary least squares where the treatment interacts with an additive combination of the group variables. Second, ordinary least squares with fully crossed interactions between the grouping variables. The fully crossed model is somewhat extreme, but it appears

regularly in applied research, especially for subgroup effects in factorial experiments.¹² Third, a hierarchical model where an additive combination of grouping variables predicts treatment effects in each group and partial pooling regularizes effect variance. In the hierarchical model, each grouping variable enters the heterogeneity equation additively, with a random intercept for each group. Fully crossing the group variables would saturate the model and make the random intercepts redundant.

There are eight data-generating processes. One uses equal group sizes and normal effect distribution, which matches the partial pooling process well. Another uses heavy tails, drawing effects from a t-distribution with three degrees of freedom. A second creates imbalanced groups with the normal effect distribution, because larger groups have more information and can thus diverge more from the overall mean, while partial pooling shrinks smaller groups closer to the overall mean. To further test this mechanism, I implement another process where larger groups have larger effects, so partial pooling takes small groups closer to a potentially unrepresentative mean. Other tests include a sparse effect distribution with only a few different estimates, a bimodal distribution, and one with systematic effect variation that might be especially friendly to the additive interactions. The last process makes effects exactly additive in the modifiers, so the additive interaction model is correctly specified.

When I divide the 2,000 observations into groups based on combinations of three to seven binary variables and give equal probability of assigning observations to a group, there are between 8 and 128 unique groups. This gives approximately 250 observations per group at the low end and 16 observations at the high end. The imbalanced simulation can have hundreds of data points in one group and as few as two in another, but I fix the data-generating process so that all groups populate. Each condition uses 200 Monte Carlo replications. For ease of computation, accuracy results come from restricted maximum likelihood fits of the hierarchical

¹²An application in the appendix by Tomz and Weeks (2021) takes this approach.

model and error rates from Bayesian estimation.¹³

I then compare the models on root-mean squared error (RMSE) and the rate of type M and type S errors. The root-mean squared error of the coefficient estimates compared to the true value is the most general metric, and is the best because the hierarchical model's shrinkage will introduce bias but compensate with lower variance (Efron and Morris, 1975; Clifford and Rainey, 2024). The bias-variance trade-off is the core of the hierarchical advantage. Lower variance may overcome greater bias in specific coefficients and make the hierarchical model more accurate on average. Pooling to the treatment mean is conservative in some ways, but that conservative approach is useful when we never know the true distribution of treatment effects. RMSE also captures instances where an estimator exaggerates the magnitude of the coefficient.

I summarize the RMSE of the hierarchical and OLS models in Figure 2. All three models perform worse as the number of groups increases and each group becomes smaller and noisier. OLS with fully crossed interactions of grouping variables performs comparably to the hierarchical model when there are 8 groups, but RMSE increases dramatically as the number of groups increases. Unequal group sizes make this problem especially acute.

The additive OLS specification does not decay in the same way. For some group configurations, there is little difference as group size increases. The downside is consistently larger RMSE relative to the hierarchical model across every group size, except for when the effects are exactly additive. Under additive heterogeneity, a correctly specified additive interaction model is 3% more accurate.

The hierarchical model is usually best in RMSE, especially as groups proliferate. With few groups, some OLS models perform marginally better, but not clearly so. Like most models, accuracy degrades as groups proliferate, because there is greater scope for bias.

These patterns are the result of reduced variance overcoming greater bias in the hierarchical

¹³In the appendix, I document that variance hyper-priors do not change the results.

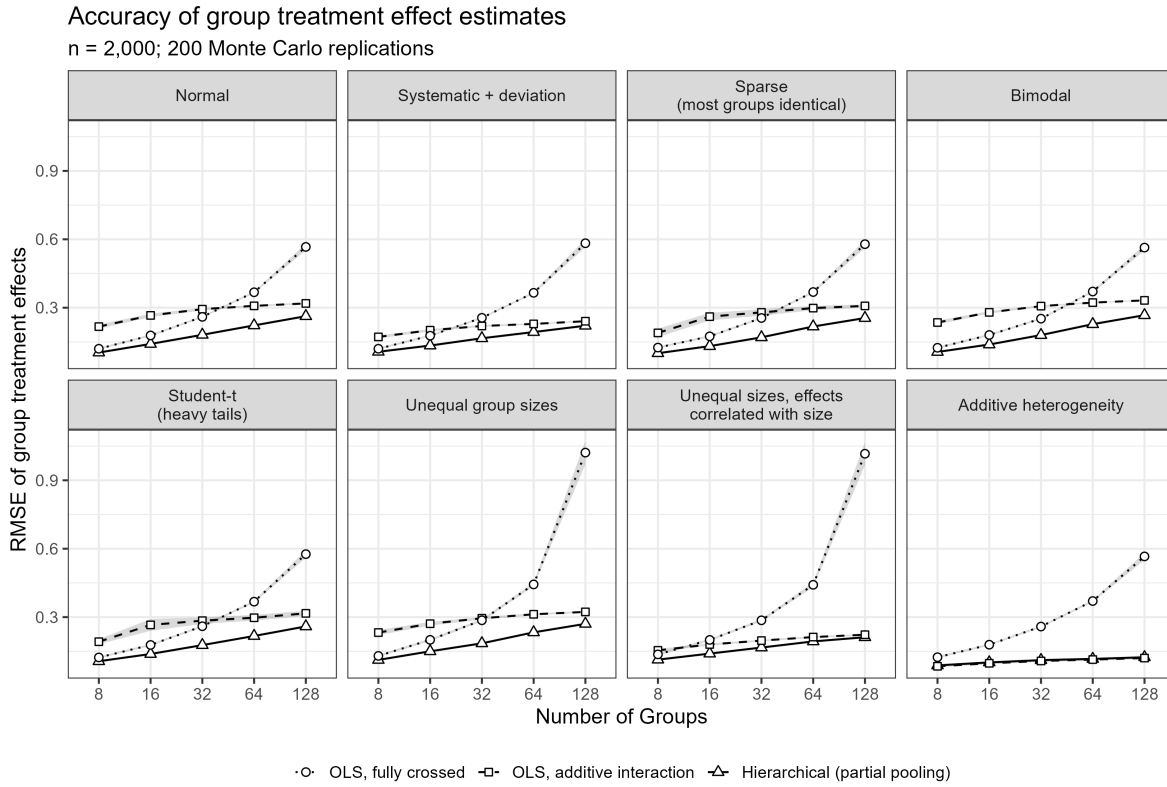


Figure 2. Comparison of OLS and hierarchical root mean squared error in group coefficient estimates. Simulation fixes the sample size at 2,000 observations and varies the number of groups. Bands give 95% Monte Carlo intervals.

estimates. I illustrate this pattern in Figure 3, which draws from the simulations with 8 and 128 groups with a normal effect distribution for ease of presentation. This figure plots the true against estimated values in those simulations for the three model specifications, using the normal distribution of true effects to illustrate. To clarify the pattern, I regress the true values on the estimated values.

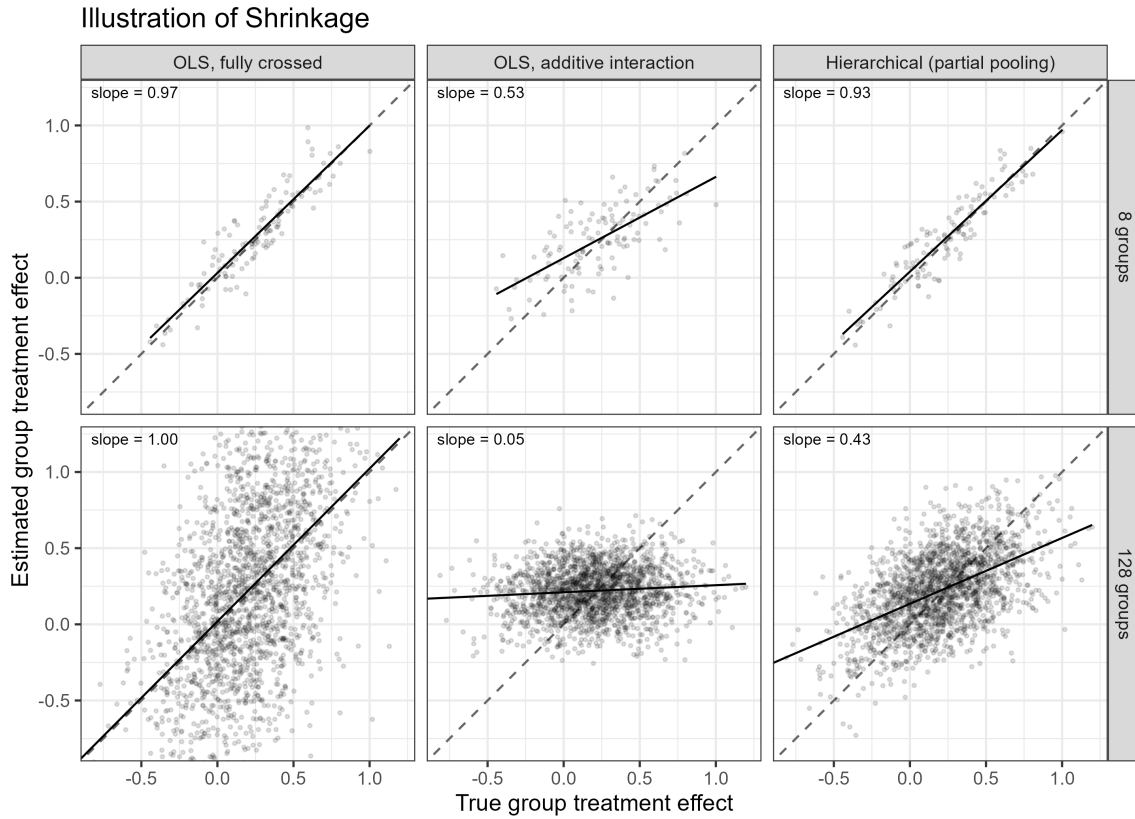


Figure 3. True and estimated effects from simulations with 8 and 128 groups. The black line regresses the true value on the estimates. Points on the dashed line are unbiased.

Under 8 groups, the hierarchical models and fully crossed OLS have similar slopes, with marginally less variance around the true values for the hierarchical model. This is why these two specifications have similar RMSE with a few groups. OLS with an additive interaction of treatment and modifiers gives a single common answer, smoothing out too much of the true variation.

At the other extreme of 128 groups, fully crossed interactions have significant noise around the true value. This variance increases RMSE, despite less bias in the overall trend. The additive model flattens out the overall trend, giving a single trend. It also has noise, but far less than the fully-crossed OLS, so the RMSE is much better. The hierarchical model falls in between. It has more bias than fully crossed OLS, but much lower variance. It recovers the true values better than the additive OLS specification, with somewhat more variation.

This is the bias-variance tradeoff in action. Hierarchical models are more biased than some specifications, but the reduction in variance makes them more accurate on average. This conservative approach is especially important when there are many modifiers and group size decreases.

Conservative estimation of subgroup effects is especially important when researchers use statistical significance thresholds. Hierarchical modeling reduces the risk that statistically significant results have the wrong sign or an exaggerated size (Gelman and Carlin, 2014). These Type S (sign) and Type M (magnitude) errors are just as important as whether an uncertainty interval includes zero. Misstating the direction of an effect misleads in theory testing, and magnitude errors fail to capture the substantive importance of an effect.

Figure 4 shows the rates of Type S and M errors in estimates where the 95% confidence or credible interval excludes zero.¹⁴ This figure plots error rates at 8, 32, and 128 groups. The rate gives the percentage of estimates with a sign error and the exaggeration ratio for Type M errors.

With sign errors, hierarchical models and fully crossed OLS perform similarly. The additive OLS model makes more sign errors because it fails to pick up adequate variation, especially as the number of groups increases. In some cases, as many as 1/3 of statistically significant effects are the wrong sign using additive OLS interactions. Other models make errors too, but they are far more rare.

¹⁴Some OLS estimates drop due to perfect collinearity and are excluded.

Errors of inference among subgroup effects declared significant

Intervals excluding zero: posterior for the hierarchical model, delta method for OLS.

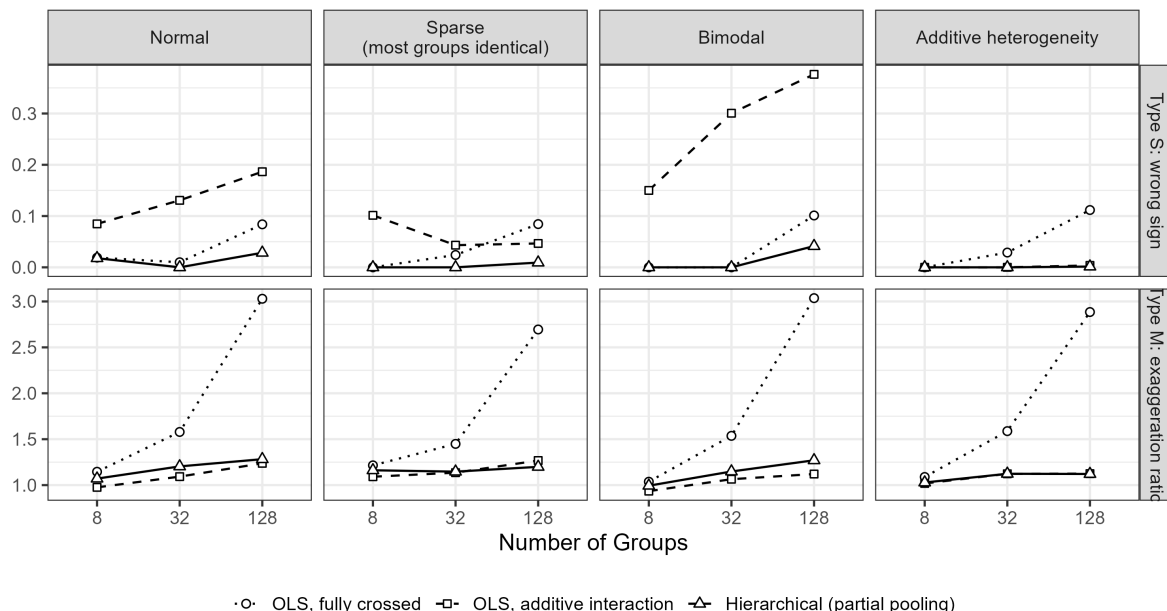


Figure 4. Type M and S error rates.

Fully crossed OLS has more problems of effect magnitude, as statistically significant effects are often large overestimates. With small groups and substantial noise, the exaggeration ratio approaches 3. Additive OLS and hierarchical models are less likely to exaggerate.

The simulation therefore shows the virtues of hierarchical modeling in three steps. In the aggregate, hierarchical models outperform the most common OLS specifications because their estimates are less variable, although slightly more biased due to partial pooling. This reduces the rate of misleading conclusions about the sign and magnitude of non-zero effects.

3 Reanalysis: Barber and Pope 2019

To further illustrate how this hierarchical approach works in practice, I reanalyze a study by Barber and Pope (2019). This clever paper attempts to disentangle ideology and party affiliation with an experiment that attributes liberal or conservative cues to Donald Trump.

They show that liberal cues from Trump move Republican attitudes in a liberal direction, while conservative cues move the same attitudes the other way.

After showing the average effects, Barber and Pope (BP henceforth) analyze how knowledge, party attachment, Trump approval and ideology modify the impact of these cues. They do this with four distinct models, one for each modifier. I unify these tests in a single hierarchical framework, which is crucial because as BP acknowledge, these four modifiers are highly correlated (Barber and Pope, 2019, pg. 50). As an additional comparison, I also fit an interaction model that enters each modifier additively. Details of the grouping decisions, diagnostics of the Bayesian model, and example code are available in the appendix.

3.1 Results

Figure 5 plots marginal effects under three different approaches: the original BP separate models, one unified interaction, and the hierarchical model. In this figure, positive values imply that a cue moves policy in a more liberal direction. Negative values imply more conservative preferences.

While BP’s four models find that each modifier changes the impact of Trump cues in a statistically significant way, unifying the estimates under a single model yields quite different results. Both the unified interaction model and the hierarchical model give less precise estimates, and usually less extreme point estimates as well. The hierarchical estimates are the most uncertain, but that additional variance is a true reflection of the correlations in these modifiers.

Most of the estimates retain the same sign, at least, or move from positive or negative to more negligible. This suggests that the direction of the relationship in the aggregate makes sense, but that this design cannot sort out their independent effect. The strong correlations between party identification, Trump approval and ideology make it hard to identify unique moderating effects.¹⁵ Political knowledge is more orthogonal, and this makes the estimates

¹⁵Given the correlations in these four modifiers, another approach might have been to use a factor analysis or

Interaction Effects: Barber and Pope 2019

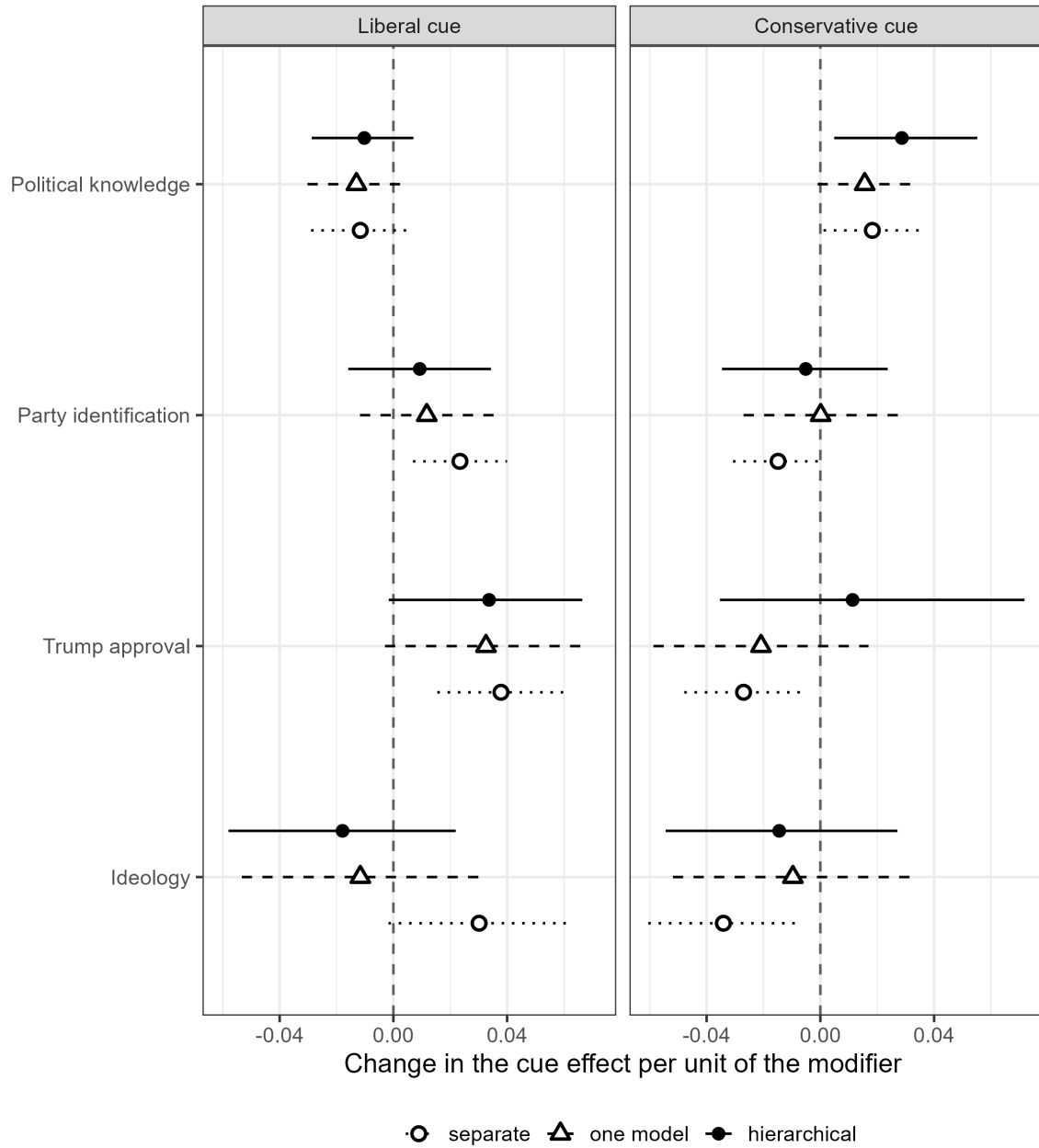


Figure 5. Estimates of the marginal effect of liberal and conservative Trump cues on policy attitudes. Error bars show the 95% confidence or credible interval.

more precise.

Perhaps as a result, only political knowledge has a clearly non-zero relationship with cue responses, once the modifier models are unified. It also moves in a somewhat unexpected direction, as conservative cues from Trump move attitudes in a more liberal direction as political knowledge increases. This is also what BP find, and they attribute it to greater party loyalty among less knowledgeable Republican voters.

The hierarchical model also provides some information about residual effect variation that the modifiers do not capture, akin to the residual in a regression model. For conservative cues, the standard deviation of the treatment effects is .1, while it is .03 for liberal cues. The four modifiers account for most of the variation in responses to liberal cues. The hierarchical model picks up much more of the variation the modifiers miss in the conservative cues analysis. Thus, the hierarchical model gives a more accurate picture of the uncertainty and explanatory power of the modifiers.

These results show some potential uses of the hierarchical approach to heterogeneous effects. Unifying disparate interactions in a single model is wise, and in doing so, scholars should also regularize those estimates. In this case, regularization makes it very clear that correlations between three of the modifiers make clear conclusions about each particular modifier unsupported.

4 Conclusion

As conditional theories proliferate in social science, many independent variables have multiple potential modifiers. Interaction models cannot easily handle this shift, as adding three or more variables to an interaction reduces the interpretation benefits and fitting separate models for each modifier increases the risk of spurious results. Hierarchical modeling retains the clear other model to capture the common concept or two behind party identification, approval and ideology.

interpretation of interactions in a single model that regularizes estimates.

I have therefore explained how to use hierarchical models to estimate heterogeneous effects. The first step is to identify sources of heterogeneity, usually unit characteristics or other treatments, and set corresponding groups. Then, fit two regressions, one that models the relationship between a treatment and the outcome, and another that models the treatment effect with the modifiers and varying intercepts by group.

Explicitly modeling how different groups respond to an independent variable can help test arguments and inform policy. It also details what drives variation in an effect and how much an effect varies. By using partial pooling of different groups, hierarchical modeling is less vulnerable to spurious inferences about subgroups.

The main downside of this model is that as a researcher adds more groups and group size shrinks, partial pooling becomes less useful. Small groups lack sufficient data to offset the prior, and are thus pooled strongly to the overall mean. If all the groups are small, this reduces the estimated variance in effect sizes so much that in extreme cases, it will collapse to basically zero. At this point, the systematic component in the regression will dominate and produce conclusions that are closer to interaction models.

Hierarchical modeling provides an intermediate approach between interactions and machine learning algorithms. For interactions with one or perhaps two modifiers, relying on simple interaction tools is best. Machine learning is best for discovery of complex heterogeneity, especially in large data. When there are two or more theoretically informed modifiers and researchers want to test their relative weight, hierarchical modeling allows flexible and interpretable estimation of effect variation.

Future work can build on this approach in other ways. Gaussian process regression is another flexible estimation tool with similar benefits that could extend to similar problems (Prati et al., 2023). Considering temporal, spatial and spatiotemporal autocorrelation in both equations is another worthwhile task. Another interesting application might be pooling experimen-

tal data and using the hierarchical structure to compare different manipulations in experiments with similar structure and smaller modifications. These tasks would further expand the benefits of hierarchical modeling in incorporating multiple conditional arguments into a single estimation framework.¹⁶

¹⁶AI Disclosure: I used Claude Code to write and refine the simulation and analysis code, including the analysis of prior publications encapsulated in figure 1. The classification itself also uses an open-weight language model (gpt-oss-120b). I also checked all the code for bugs or problems with Claude Code.

References

- Alley, Joshua. 2021. "Alliance Participation, Treaty Depth and Military Spending." *International Studies Quarterly* 65(4):929–943.
- Arel-Bundock, Vincent, Ryan C Briggs, Hristos Doucouliagos, Marco Mendoza Aviña and Tom D Stanley. 2026. "Quantitative Political Science Research Is Greatly Underpowered." *The Journal of Politics* 88(1):36–46.
- Barber, Michael and Jeremy C Pope. 2019. "Does Party Trump Ideology? Disentangling Party And Ideology in America." *American Political Science Review* 113(1):38–54.
- Barnhart, Joslyn N, Robert F Trager, Elizabeth N Saunders and Allan Dafoe. 2020. "The Suffragist Peace." *International Organization* 74(4):633–670.
- Blackwell, Matthew and Michael P Olson. 2022. "Reducing Model Misspecification and Bias in the Estimation of Interactions." *Political Analysis* 30(4):495–514.
- Brambor, Thomas, William Roberts Clark and Matt Golder. 2006. "Understanding interaction models: Improving empirical analyses." *Political Analysis* 14(1):63–82.
- Bürkner, Paul-Christian. 2017. "brms: An R package for Bayesian multilevel models using Stan." *Journal of Statistical Software* 80(1):1–28.
- Chaudoin, Stephen. 2014. "Promises or Policies? An Experimental Analysis of International Agreements and Audience Reactions." *International Organization* 68(1):235–256.
- Clark, William Roberts and Matt Golder. 2023. *Interaction Models: Specification and Interpretation*. Cambridge University Press.
- Clifford, Scott and Carlisle Rainey. 2024. "Estimators for Topic-Sampling Designs." *Political Analysis* 32(4):431–444.
- Davies, Graeme AM and Robert Johns. 2013. "Audience costs among the British public: the impact of escalation, crisis type, and prime ministerial rhetoric." *International Studies Quarterly* 57(4):725–737.
- Dorie, Vincent, George Perrett, Jennifer L Hill and Benjamin Goodrich. 2022. "Stan and BART for Causal Inference: Estimating Heterogeneous Treatment Effects Using the Power of Stan and the Flexibility of Machine Learning." *Entropy* 24(12):1782.
- Efron, Bradley and Carl Morris. 1975. "Data analysis using Stein's estimator and its generalizations." *Journal of the American Statistical Association* 70(350):311–319.
- Feller, Avi and Andrew Gelman. 2015. "Hierarchical Models for Causal Effects." *Emerging Trends in the Social and Behavioral Sciences: An interdisciplinary, searchable, and linkable resource* pp. 1–16.

- Gelman, Andrew. 2018. "You need 16 times the sample size to estimate an interaction than to estimate a main effect." Available at: <https://statmodeling.stat.columbia.edu/2018/03/15/need16/>.
- Gelman, Andrew and Jennifer Hill. 2007. *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Vol. 1 Cambridge University Press New York, NY, USA.
- Gelman, Andrew and John Carlin. 2014. "Beyond Power Calculations: Assessing Type S (Sign) and Type M (Magnitude) Errors." *Perspectives on Psychological Science* 9(6):641–651.
- Goplerud, Max. 2021. "Modelling Heterogeneity Using Bayesian Structured Sparsity." *arXiv preprint arXiv:2103.15919*.
- Green, Donald P and Holger L Kern. 2012. "Modeling Heterogeneous Treatment Effects in Survey Experiments with Bayesian Additive Regression Trees." *Public Opinion Quarterly* 76(3):491–511.
- Grimmer, Justin, Solomon Messing and Sean J Westwood. 2017. "Estimating Heterogeneous Treatment Effects and the Effects of Heterogeneous Treatments with Ensemble Methods." *Political Analysis* 25(4):413–434.
- Guisinger, Alexandra and Elizabeth N. Saunders. 2017. "Mapping the Boundaries of Elite Cues: How Elites Shape Mass Opinion across International Issues." *International Studies Quarterly* 61(2):425–441.
- Hahn, P Richard, Carlos M Carvalho, David Puelz and Jingyu He. 2018. "Regularization and confounding in linear regression for treatment effect estimation." *Bayesian Analysis* 13(1):163–182.
- Imai, Kosuke and Marc Ratkovic. 2013. "Estimating treatment effect heterogeneity in randomized program evaluation." *The Annals of Applied Statistics* 7(1):443–470.
- Kertzer, Joshua D and Ryan Brutger. 2016. "Decomposing Audience Costs: Bringing the Audience Back into Audience Cost Theory." *American Journal of Political Science* 60(1):234–249.
- Künzel, Sören R, Jasjeet S Sekhon, Peter J Bickel and Bin Yu. 2019. "Metalearners for estimating heterogeneous treatment effects using machine learning." *Proceedings of the national academy of sciences* 116(10):4156–4165.
- Levendusky, Matthew S and Michael C Horowitz. 2012. "When Backing Down is the Right Decision: Partisanship, New Information, and Audience Costs." *The Journal of Politics* 74(2):323–338.
- Levy, Jack S, Michael K McKoy, Paul Poast and Geoffrey PR Wallace. 2015. "Backing Out or Backing in? Commitment and Consistency in Audience Costs Theory." *American Journal of Political Science* 59(4):988–1001.

- Lin-Greenberg, Erik. 2019. "Backing up, not backing down: Mitigating audience costs through policy substitution." *Journal of Peace Research* 56(4):559–574.
- Marquardt, Kyle L. 2022. "Language, Ethnicity, and Separatism: Survey Results from Two Post-Soviet Regions." *British Journal of Political Science* 52(4):1831–1851.
- McElreath, Richard. 2016. *Statistical Rethinking: A Bayesian course with examples in R and Stan*. CRC Press.
- Much, Melina. 2025. "Bayesian multilevel modeling for the intersections of race, gender, and class." *Political Research Quarterly* 78(3):985–1006.
- Prati, Annamaria, Yehu Chen, Jacob Montgomery and Roman Garnett. 2023. "A Gaussian Process Framework for Structured, Flexible, and Interpretable Machine-Learning Models."
- Ratkovic, Marc and Dustin Tingley. 2017. "Sparse estimation and uncertainty with application to subgroup analysis." *Political Analysis* 25(1):1–40.
- Schwartz, Joshua A and Christopher W Blair. 2020. "Do Women Make More Credible Threats? Gender Stereotypes, Audience Costs, and Crisis Bargaining." *International Organization* 74(4):872–895.
- Simmons, Joseph P, Leif D Nelson and Uri Simonsohn. 2011. "False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant." *Psychological Science* 22(11):1359–1366.
- Stegmuller, Daniel. 2013. "How many countries for multilevel modeling? A comparison of frequentist and Bayesian approaches." *American Journal of Political Science* 57(3):748–761.
- Tomz, Michael. 2007. "Domestic Audience Costs in International Relations: An Experimental Approach." *International Organization* 61(4):821–40.
- Tomz, Michael and Jessica L.P. Weeks. 2021. "Military Alliances and Public Support for War." *International Studies Quarterly* 65(3):811–824.
- Wager, Stefan and Susan Athey. 2018. "Estimation and Inference of Heterogeneous Treatment Effects using Random Forests." *Journal of the American Statistical Association* 113(523):1228–1242.